Experimental & Behavioral Economics, Spring 2024

Course Number: Econ 689

Instructor

Professor David Gill
Email: gill53@purdue.edu
Office: Rawls 4023

Course description

This half-semester graduate module is intended to prepare graduate students for research in experimental economics. The module will be open to graduate students in economics and other related fields such as management, marketing, finance and engineering / computer science. The module can be taken alone. The module can also be taken before or after ECON 685 and ECON 686, and will cover different topics from those modules.

The course will introduce students to the methodology of experimental economics and describe statistical techniques for analyzing experimental data. The course will also cover a number of specific research topics, using a mixture of lecture materials and class discussion of seminal research papers. These topics will intersect with behavioral economics, which aims to identify empirically important deviations from traditional microeconomic assumptions that help to explain economic behavior and guide policymakers. Finally, the course will instruct students in how to present research effectively in research seminars and in papers to be submitted for publication.

Intended learning outcomes

- Develop skills to design experiments that test cleanly predictions from economic theory.
- Understand statistical techniques appropriate for different types of experimental designs.
- Become familiar with cutting-edge research in the field of experimental and behavioral economics.
- Gain experience in how to present research effectively.
- Learn how to write research papers that will appeal to the top journals in the field.

Library Resources

- The main text is: Dhami, S. (2016). The Foundations of Behavioral Economic Analysis, Oxford University Press
- You can access an electronic copy through Purdue Libraries

Enrollment:

Between 5 and 25 students.

Provisional Outline

Topic 1. Introduction to the methodology of experiments
Topic 2. Some statistical considerations in experiments
Topic 3. Introduction to behavioral game theory
- Topic 3.1: Psychological games
- Topic 3.2: Experimental behavior in simple games
- Topic 3.3: Level-k

Topic 4. Loss aversion
- Topic 4.1: Prospect theory
- Topic 4.2: Expectations based reference points
- Topic 4.3: Goals based reference points

Topic 5. Cognitive skills

Evaluation

A & B are individual assignments

C & D are group assignments
- **You will form groups of three**
- To do this, you can use the discussion forum on Brightspace (Course tools -> Discussions)
- Each group will prepare two paper presentations (C) and one power analysis presentation (D)
- All three group members should contribute equally to each of these assignments
- **Each group member will present one of the assignments in class**
- That is, one group member will present the first paper presentation, the second will present the second paper presentation, and the third will present the power analysis
- For each of the three assignments, the same percentage grade will be allocated to all three group members

A. *General class participation* [20%]
- You will be expected to ask questions and to participate actively in class discussions.
- **When another student presents a paper, you must read the paper in detail in advance of class.** You will be expected to ask good questions, make insightful comments, and be able to summarize and explain key aspects of the paper (theoretical framework, experimental design, results).

B. *Write an introduction for an existing paper* [30%] [individual assignment]
- **Due on 04/29 (Monday of finals week)**
- Choose an experimental paper published in a top-five economics journal[1] in the last ten years that: (i) you have not read or seen presented; and (ii) is on a topic that you have not studied in any depth.
- Read the sections on experimental design, theoretical framework (if any) and results.

[1] American Economic Review, Econometrica, Journal of Political Economy, Quarterly Journal of Economics, Review of Economic Studies

- Do not read the introduction, literature review, discussion / conclusion, or list of references (you can read the abstract).
- Write a full introduction to the paper, including how the paper relates to the existing literature (maximum of 5 pages).
- After you have finished writing your introduction, compare your introduction to the actual introduction, and write a critique of your introduction in light of the actual one.

C. *Two paper presentations* [15% each] [group assignment]
- From the papers marked [*] in the reading list.
- The presentation should be ordered as follows: (i) theoretical framework; (ii) experimental design; (iii) econometric framework and results; (iv) relationship to the existing literature; (v) critique / comments on the paper.
- **The presentation should last 30mns** (allowing for questions; we will then discuss the paper for c. 15mns).
- **By end of Monday 03/18 each group should email me their preference ordering over the six papers marked [*]**

D. *Power analysis presentation* [20%] [group assignment]
- 1. Take an existing experimental economics paper from the literature that: (i) has a clean between-subject treatment-control comparison; and (ii) finds a treatment-control difference at the 5% significance level.
- 2. Pretend that the experiment has not yet been run (so do not look at the data in any detail).
- 3. Build a model of the data-generating process (DGP) in the treatment and in the control (when doing this, think carefully about how you model variance/noise in behavior; if possible use existing literature to inform your assumptions).
- 4. Assuming that the average treatment effect in your DGP is the estimated treatment effect from the experiment, simulate many datasets (each of the same sample size as the sample in the experiment).
- 5. Using the 5% significance criterion, use your simulations to estimate the *power* of a t-test given your DGP and sample size
- 6. Now forget your DGP, but remember the power that you estimated. For each of three different assumptions about prior beliefs, calculate the *post-study probability (PSP)* for the experimental finding that you have chosen.
- 7. Repeat steps 4 and 5, assuming now that the average treatment effect in your DGP is half of the estimated treatment effect from the experiment.
- 8. Repeat steps 4 and 5, assuming now that the sample size is twice the sample size used in the experiment.
- 9. Discuss any lessons that your analysis suggests for how to design experiments.
- 10. **The presentation should last 40mns.**

Use of Artificial Intelligence (AI)

- AI tools can help learning and understanding.
- In this course you are allowed to use AI tools to help learning and understanding, in the same way that you use the material in the textbook or lecture slides.
- When using AI tools to aid learning, remember that the work that you submit must remain your own. In particular, you must not pass off the output of AI tools as your own work

Readings

The main readings are listed below. They include some of the seminal articles on each topic together with survey pieces and textbook treatments. **For some topics, but not all, I will provide additional references in the lecture notes.**

Topic 1: Introduction to the methodology of experiments

1.1 Methodology

Dhami, S. (2016). The Foundations of Behavioral Economic Analysis, Oxford University Press, Section 3 of the Introduction and Chapter 5.5

Bardsley, Nicholas et al. (2009). Experimental Economics: Rethinking the Rules, Princeton University Press

Camerer, C.F. (2003). Behavioral Game Theory: Experiments in Strategic Interaction, Princeton University Press, Section A1.2

Falk, A. and Heckman, J.J. (2009). Lab experiments are a major source of knowledge in the social sciences. Science, 326(5952): 535–538

Gill, D. and Prowse, V. (2019). Measuring costly effort using the slider task. Journal of Behavioral and Experimental Finance, 21, 1-9

1.2 Applications

Bartling, B., Fehr, E., and Schmidt, K.M. (2012). Screening, competition, and job design: Economic origins of good jobs. American Economic Review, 102(2): 834–864

Gill, D., Kissova, K., Lee, J., and Prowse, V. (2019). First-place loving and last-place loathing: How rank in the distribution of performance affects effort provision. Management Science, 65(2), 494-507

Topic 2: Some statistical considerations in experiments

2.1 Methodology

Czibor, E., Jimenez-Gomez, D., and List, J.A. (2019). The Dozen Things Experimental Economists Should Do (More of). Southern Economic Journal, 86.2: 371-432.

List, J.A., Sadoff, S., and Wagner, M. (2011). So you want to run an experiment, now what? Some simple rules of thumb for optimal experimental design. Experimental Economics, 14(4): 439–457

Maniadis, Z., Tufano, F. and List, J.A. (2014). One swallow doesn't make a summer: New evidence on anchoring effects. American Economic Review, 104(1): 277-290

Page, L., Noussair, C. N. and Slonim, R. (2021). The replication crisis, the rise of new research practices and what it means for experimental economics. Journal of the Economic Science Association, 7, 210-225.

2.2 Application

Camerer, C.F. et al. (2016). Evaluating replicability of laboratory experiments in economics. Science, 351(6280): 1433-1436.

Topic 3: Introduction to behavioral game theory

3.1 Psychological games

Battigalli, P. and Dufwenberg, M. (2022). Belief-dependent motivations and psychological game theory. Journal of Economic Literature, 60(3), 833-882.

Dhami, S. (2016). The Foundations of Behavioral Economic Analysis, Oxford University Press, Chapter 13.5

Charness, G. and Dufwenberg, M. (2006). Promises and partnership. Econometrica, 74(6): 1579-1601

[*] Dufwenberg, M., Johansson-Stenman, O., Kirchler, M., Lindner, F. and Schwaiger, R. (2022). Mean markets or kind commerce?. Journal of Public Economics, 209, 104648.

3.2 Experimental behavior in simple games

Camerer, C.F. (2003). Behavioral Game Theory: Experiments in Strategic Interaction, Princeton University Press, Chapters 1-3

Dhami, S. (2016). The Foundations of Behavioral Economic Analysis, Oxford University Press, Chapter 5.2 and 12.3

Engel, C. (2011). Dictator games: A meta study. Experimental Economics, 14(4): 583–610

Walker, M. and Wooders, J. (2001). Minimax play at Wimbledon. American Economic Review, 91(5): 1521–1538

3.3 Level-k

Camerer, C.F. (2003). Behavioral Game Theory: Experiments in Strategic Interaction, Princeton University Press, Chapter 5.2 and 5.6

Nagel, R. (1995). Unraveling in guessing games: An experimental study. American Economic Review, 85(5): 1313-1326

[*] Camerer, C.F., Ho, T.H. and Chong, J.K. (2004). A cognitive hierarchy model of games. Quarterly Journal of Economics, 119(3): 861-898

Costa-Gomes, M.A. and Crawford, V.P. (2006). Cognition and behavior in two-person guessing games: An experimental study. American Economic Review, 96(5): 1737-1768

Crawford, V.P., Costa-Gomes, M.A. and Iriberri, N. (2013). Structural models of nonequilibrium strategic thinking: Theory, evidence, and applications. Journal of Economic Literature, 51(1): 5-62

Dhami, S. (2016). The Foundations of Behavioral Economic Analysis, Oxford University Press, Chapter 13.3 and 13.4

<u>Topic 4: Loss aversion</u>

4.1 Prospect theory

Dhami, S. (2016). The Foundations of Behavioral Economic Analysis, Oxford University Press, Chapter 1.5.4, 2.4, 2.5 and 3.6 (**especially pp.130-132 that explains the value function**)

Kahneman, D. and Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. Econometrica, 47(2): 263-292

Rabin, M. (2000). Risk aversion and expected-utility theory: A calibration theorem. Econometrica, 68(5):1281-1292

Tversky, A. and Kahneman (1992). Advances in prospect theory: Cumulative representation of uncertainty. Journal of Risk and Uncertainty. 5(4): 297-323

4.2 Expectations based reference points

Dhami, S. (2016). The Foundations of Behavioral Economic Analysis, Oxford University Press, Chapters 2.8 & 3.7

Kőszegi, B. and Rabin, M. (2006). A model of reference-dependent preferences. Quarterly Journal of Economics, 121(4): 1133-1165

[*] Abeler, J., Falk, A., Goette, L. and Huffman, D. (2011). Reference points and effort provision. American Economic Review, 101(2): 470-492

[*] Gill, D. and Prowse, V. (2012). A structural analysis of disappointment aversion in a real effort competition. American Economic Review, 102(1): 469-503

Gill, D. and Stone, R. (2010). Fairness and desert in tournaments. Games and Economic Behavior, 69(2), 346-364

Card, D. and Dahl, G.B. (2011). Family violence and football: The effect of unexpected emotional cues on violent behavior. Quarterly Journal of Economics, 126(1): 103-143

Pagel, M. (2017). Expectations-Based Reference-Dependent Life-Cycle Consumption. Review of Economic Studies. 84: 885–934

Campos-Mercade, P., Goette, L., Graeber, T., Kellogg, A. and Sprenger, C. (2023). "De Gustibus" and Disputes about Reference Dependence. HBS Working Paper 24-046

4.3 Goals based reference points

Koch, A.K. and Nafziger, J. (2011). Self-regulation through Goal Setting. Scandinavian Journal of Economics, 113(1): 212-227

Corgnet, B., Gómez-Miñambres, J. and Hernán-Gonzalez, R. (2015). Goal setting and monetary incentives: When large stakes are not enough. Management Science, 61(12): 2926-2944

Clark, D., Gill, D., Prowse, V.L. and Rush, M. (2020). Using Goals to Motivate College Students: Theory and Evidence from Field Experiments. Review of Economics and Statistics, 102(4): 648-663

Topic 5: Cognitive skills

[*] Gill, D. and Prowse, V. (2016). Cognitive ability, character skills, and learning to play equilibrium: A level-k analysis. Journal of Political Economy, 124(6): 1619-1676

Alaoui, L. and Penta, A. (2016). Endogenous depth of reasoning. Review of Economic Studies, 83(4), 1297-1333.

Proto, E., Rustichini, A. and Sofianos, A. (2019). Intelligence, personality, and gains from cooperation in repeated interactions. Journal of Political Economy, 127(3), 1351-1390.

Alaoui, L., Janezic, K. A. and Penta, A. (2020). Reasoning about others' reasoning. Journal of Economic Theory, 189, 105091.

[*] Weidmann, B. and & Deming, D. (2021). Team players: How social skills improve team performance. Econometrica, 89(6), 2637-2657.

Fe, E., Gill, D. and Prowse V. (2022). Cognitive skills, strategic sophistication, and life outcomes. Journal of Political Economy, 130(10), 2643-2704.

Gill, D. and Prowse, V. (2023). The creativity premium: Exploring the link between childhood creativity and life outcomes. Journal of Political Economy Microeconomics, forthcoming. https://web.ics.purdue.edu/~gill53/GillProwse_Creativity.pdf

MobLab

- We will use MobLab to run in-class experiments.
- We will access the MobLab interface using the internet.
- Please bring a computer or smartphone to class to access MobLab.
- Please create a free Moblab account: see next page for instructions.
- You must use your university email address to register
- If you are using a smartphone, then download the app (see next page).

MobLab
A playground for decisions

3219 E Foothill Blvd
Pasadena, CA 91107

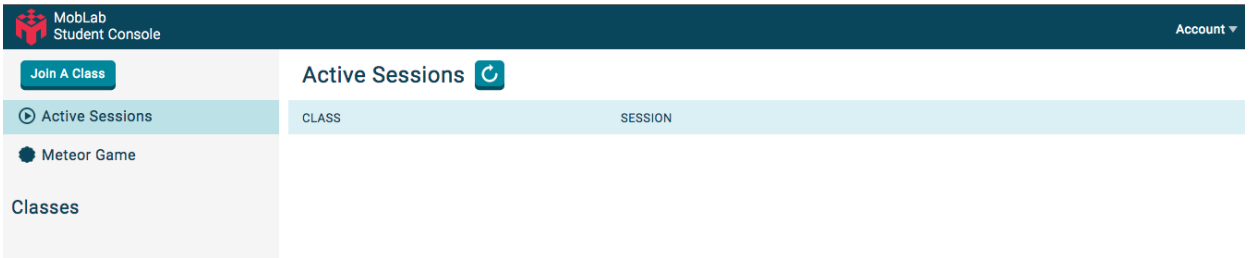
(Tel): 626-793-5070
(Web): www.moblab.com

Register with MobLab

To facilitate recordkeeping, I ask that you use your university email address. You can register using any browser (https://game.moblab.com/server/register.jsp). To subsequently access the Student Console, log in at https://game.moblab.com/server/login.

Join the class

In any browser or in the MobLab app, access the Student Console (see Figure 1). Choose Join a Class, and enter our Class Code: **x23kkb524**



In the resulting popup, you should have confirmation that you've joined my **ECON689_24** class.

What do I need to participate?

To participate in MobLab games you need an internet-connected device. Laptop users can use any browser. **I strongly** recommend that iOS (iPhone and iPad) and Android users use the MobLab app downloadable from the device's app store. Trust me, the app is so much better for playing on handheld devices!

Playing games

When instructed, access the Student Console in your browser or open the app on your iOS or Android device. You will see your class's session if your instructor has already started it. If not, wait for your instructor to start the session and click the refresh button ( ). Once the session is visible, click Join (web browser) or the session name (Android/iOS app).

If you have any questions or run into any issues, please email support@moblab.com

GENERAL POLICIES

Computers and cell phones

Students may use computers and cellphones during class for academic purposes only (e.g., note-taking). Web surfing and other non-academic activities are highly disruptive to other students and are prohibited.

Computers and cellphones must be turned to silent before class begins.

Attendance

Students are expected to attend all classes in-person unless they are ill or otherwise unable to attend class. Classroom engagement is extremely important and associated with your overall success in the course.

Only the instructor can excuse a student from a course requirement or responsibility. When conflicts can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency conflicts, when advance notification to an instructor is not possible, the student should contact the instructor by email. In cases of bereavement, quarantine, or isolation, the student or the student's representative should contact the Office of the Dean of Students via email (odos@purdue.edu) or phone at 765-494-1747.

Purdue's Grief Absence Policy for Students can be viewed here:
http://www.purdue.edu/studentregulations/regulations_procedures/classes.html

Mental Health and Wellness

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack. Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at evans240@purdue.edu.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-

6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Accessibility and disability

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Discrimination

Purdue University does not condone and will not tolerate discrimination against any individual on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, disability, status as a veteran, marital status, parental status, sexual orientation, gender identity or gender expression. Purdue University promulgates policies and programs to ensure that all persons have equal access to its employment opportunities and educational programs, services and activities.

A hyperlink to Purdue's full Nondiscrimination Policy Statement is here:
https://www.purdue.edu/purdue/ea_eou_statement.php

Purdue's Honor Pledge

"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue." This applies to all problem sets, exams, and interactions with other students and your professors.

Academic Dishonesty

Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Part 5, Section III-B-2-a, Student Regulations] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972]

Incidents of academic misconduct in this course will be addressed by the course instructor and referred to the Office of Student Rights and Responsibilities (OSRR) for review at the university level. Any academic misconduct will result minimally in a failing or zero grade for that particular assignment, and at the instructor's discretion may result in a failing grade for the course. In

addition, all incidents of academic misconduct will be forwarded to OSRR, where university penalties, including removal from the university, may be considered.

Emergencies

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website on Brightspace and/or circulated via email. You are expected to read your @purdue.edu email on a frequent basis.

Emergency preparedness

As we begin this semester I want to take a few minutes and discuss emergency preparedness. Purdue University is a very safe campus and there is a low probability that a serious incident will occur here at Purdue. However, just as we receive a "safety briefing" each time we get on an aircraft, we want to emphasize our emergency procedures for evacuation and shelter in place incidents. Our preparedness will be critical IF an unexpected event occurs!

Emergency preparedness is your personal responsibility. Purdue University is actively preparing for natural disasters or human-caused incidents with the ultimate goal of maintaining a safe and secure campus. Let's review the following procedures:

- For any emergency call 911.
- There are nearly 300 Emergency Telephone Systems throughout campus that connect directly to the Purdue Police Department (PUPD). If you feel threatened or need help, push the button and you will be connected to the PUPD.
- If we hear a fire alarm we will immediately evacuate the building and proceed to the grass in front of the Union across the street from Krannert
    - **Do not use the elevator.**
- If we are notified of a Shelter in Place requirement for a tornado warning we will shelter in the lowest level of this building away from windows and doors. Our preferred location is the lower lever (basement) or tunnel.
- If we are notified of a Shelter in Place requirement for a hazardous materials release we will shelter in our classroom shutting any open doors and windows.
- If we are notified of a Shelter in Place requirement for a civil disturbance such as a shooting we will shelter in a room that is securable preferably without windows. Our preferred location is our classroom. We will lock the room, turn off the lights, and get under/behind the desk counters.

Disclaimer
This syllabus is subject to change.

EMERGENCY PREPAREDNESS SYLLABUS ATTACHMENT

EMERGENCY NOTIFICATION PROCEDURES are based on a simple concept – if you hear a fire alarm inside, proceed outside. If you hear a siren outside, proceed inside.

- **Indoor Fire Alarms** mean to stop class or research and immediately **evacuate** the building.
  - Proceed to your Emergency Assembly Area away from building doors. **Remain outside** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.
- **All Hazards Outdoor Emergency Warning Sirens** mean to immediately seek shelter (**Shelter in Place**) in a safe location within the closest building.
  - "Shelter in place" means seeking immediate shelter inside a building or University residence. This course of action may need to be taken during a tornado, a civil disturbance including a shooting or release of hazardous materials in the outside air. Once safely inside, find out more details about the emergency*. **Remain in place** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

**In both cases, you should seek additional clarifying information by all means possible…Purdue Emergency Status page, text message, Twitter, Desktop Alert, Albertus Beacon, digital signs, email alert, TV, radio, etc.…review the Purdue Emergency Warning Notification System multi-communication layers at http://www.purdue.edu/ehps/emergency_preparedness/warning-system.html*

EMERGENCY RESPONSE PROCEDURES:

- Review the **Emergency Procedures Guidelines** https://www.purdue.edu/emergency_preparedness/flipchart/index.html
- Review the **Building Emergency Plan** (available on the Emergency Preparedness website or from the building deputy) for:
  - evacuation routes, exit points, and emergency assembly area
  - when and how to evacuate the building.
  - shelter in place procedures and locations
  - additional building specific procedures and requirements.

EMERGENCY PREPAREDNESS AWARENESS VIDEOS

- "Shots Fired on Campus: When Lightning Strikes," is a 20-minute active shooter awareness video that illustrates what to look for and how to prepare and react to this type of incident. See: http://www.purdue.edu/securePurdue/news/2010/emergency-preparedness-shots-fired-on-campus-video.cfm (Link is also located on the EP website)

MORE INFORMATION

Reference the Emergency Preparedness web site for additional information: https://www.purdue.edu/ehps/emergency_preparedness/